

offer a 2W power rating in just a 2512 case size for use in low-noise high-precision control systems in military, aerospace, telecommunication and industrial applications. The resistors are 100 per cent visually inspected as per MIL-PRF-55342 standard and are sulphur-resistant according to ASTM B890-95 humid-vapour test.

Replacement of batteries by ultracapacitors is something that is happening in several segments such as wind turbines, vehicles, meters, engine cranking and the like. "The only issue was price, which was being addressed by manufacturers. Korea is playing a very important role in providing quality capacitors; Vinatex is one example here. However, the tipping point that caused people to move from using batteries to ultracapacitors is driven by the battery maintenance and replacement cost. Lifetime cost is also lower since ultracapacitors last much longer, while you need to change batteries frequently," says Puneet Shukla, CEO, Saison Components.

Cut down on size, raise power-handling capability

Enabling higher power capability at smaller component sizes is one of the key challenges that manufacturers have tried to solve. Vishay has launched a series named TNPV e3 that helps reduce component count and therefore save circuit space by replacing larger resistors or multiple devices in a large case.

Relation between cost and quality

Size reduction in passives is steady but because the cost per component is very low, it contributes a lower percentage of the bill-of-materials cost of the product. The printed circuit board (PCB) cost tends to be much higher. This is why going to smaller passives that cost higher would be a better idea, as PCB cost reduction has a bigger share.

Guidelines.

1. Specifications are very important, and it is understood that passives are available the cheapest from China but people do go for branded ones.

2. If the buyer just negotiates the cost, he or she might not see that the back-end and infrastructure is very important. If people are looking for quality components, they should visit the factory and see the people behind the project. Factories that do not have enough infrastructure cannot deliver on quality.

3. Qualifications of the company by infrastructure and its future plans are also important. The question you should ask is: Is it just about cost or is it a professional company or specification? Quality checks are an important area of how some companies try to cut down on cost.

Who is to be blamed? It could be us. After all, we buy from China because it is cheaper. There are good quality products from China if you pay the right price. Sunlord is an example of a Chinese company where they have American or German standards in their factory itself.

Parting thought: It is unfortunate that we are forced to source from outside India for such small components.

—Puneet Shukla, CEO, Saison Components

"Resistor size is one area of focus for our firm. The smaller sizes available today in leaded and shunt resistors are 20mm. These now come with better anti-surge features with metal housings that help release heat. These are easier to mount too, due to the fact that leads are on the same side of the chip," said Sei Komiya from Japan Resistor Manufacturing Co., during an interaction at an electronics industry event in Delhi earlier this year.

"In recent years the electronic components industry has observed a shift toward passive components that are smaller, more reliable and more powerful. Distributors and manufac-

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turers alike are investing heavily in the development of these features. Higher-density resistors, higher-current magnetics and polymer and hybrid capacitors have taken centre-stage this year, as leading passive manufacturers like Vishay, Panasonic, Ohmite, Kemet, AVX and Bourns are leveraging new techniques and technologies that change the way passive components fulfil the design needs of engineers," says Honnavar.

Some vendors are bringing out components having aluminium-nitride substrates with enlarged terminations that ease thermal conductivity between the top part of the resistor and its solder joint on the circuit board. This allows much higher power-handling capability for those resistors. Honnavar adds that technologies like device embedded substrates (DESeS) use advanced printing

SMD Resistor Sizes

Code		Length (l)		Width (w)		Height (h)		Power
Imperial	Metric	inch	mm	inch	mm	inch	mm	watt
0201	0603	0.024	0.6	0.012	0.3	0.01	0.25	1/20 (0.05)
0402	1005	0.04	1.0	0.02	0.5	0.014	0.35	1/16 (0.062)
0603	1608	0.06	1.55	0.03	0.85	0.018	0.45	1/10 (0.10)
0805	2012	0.08	2.0	0.05	1.2	0.018	0.45	1/8 (0.125)
1206	3216	0.12	3.2	0.06	1.6	0.022	0.55	1/4 (0.25)
1210	3225	0.12	3.2	0.10	2.5	0.022	0.55	1/2 (0.50)
1218	3246	0.12	3.2	0.18	4.6	0.022	0.55	1
2010	5025	0.20	5.0	0.10	2.5	0.024	0.6	3/4 (0.75)
2512	6332	0.25	6.3	0.12	3.2	0.024	0.6	1